Ghost Data Audit: Interview Cheat Sheet

> **Pitch (say this first):** "Northstar's dashboard says customers are deleted, but copies survive in replicas, search indexes, exports and backups. That's ghost data. Our agent reads all 94 evidence files, works out who the person is, including old emails, and uses only what was known at the audit time. It then applies the 15 deletion rules and outputs COMPLETE, INCOMPLETE or NOT_APPLICABLE with a cited fix plan. Two independent checkers verify every row. The final run makes zero AI calls; AI only helps where reading is fuzzy."

The 8-step flow (explain why for each)

1. **Read all files into facts**, each tagged with its source file. *Why:* every answer must cite real evidence.
2. **Intent:** did they ask for deletion? If not, NOT_APPLICABLE. *Why:* "fix my email" must never delete anything.
3. **Identity:** current plus old emails and phones. *Why:* old systems keep old identifiers.
4. **Collect** every record linked to the person.
5. **Time cutoff:** only evidence from at or before the audit time. *Why:* no seeing the future.
6. **Policy:** the rules file decides delete, redact, anonymize or keep. *Why:* deleting what must be kept is also wrong.
7. **Plan:** stop syncs and restores first, then fix, then verify. *Why:* otherwise the data comes back.
8. **Double-check:** a simulator plus a validator. *Why:* catch our own mistakes before output.

Why not just use an LLM?

Graders check exact steps and evidence IDs. LLMs can invent IDs and vary between runs. Rules are repeatable, cheap and cited. **AI is used for:** (1) classifying unclear documents, where two models (gpt-5.5 and gpt-5.4-mini) must agree and the answer is saved for offline reuse; (2) an advisory reviewer that never changes answers.

Key example (case_16)

The log says search_1007 was deleted: "SUCCESS". The database snapshot at audit time still has it, so it's INCOMPLETE: report it, delete it, verify it's gone, and keep the security log. **A claim isn't proof.**

Story / biggest lesson

Codex's version refused to answer 442 of 518 cases. "Unresolved" didn't mean missing data; our code was over-strict. Each assumption was fixed only when the 32 answer-key samples proved it wrong: **442 → 207 → 5 → 0.**

How built

Codex built the base. Claude Code then ran 5 agents in parallel, each with memory: researcher, implementer, QA, reviewer, output. A change was accepted only if the scores improved with no regressions.

Numbers to remember

Samples (answer key)	31/32 statuses correct · 20/32 exact · precision 0.88 · recall 0.82
Most dangerous errors	0 wrong "COMPLETE" · 0 wrongful deletions
Robustness	1,024 shuffled variants: answers never changed · 225 tests · 28/30 edge cases
Full run	518 rows, 0 errors · 0 AI calls at prediction time

Tough questions

- **Weakness?** 128 rows lack an account check because the data never links the account. Every pattern we tried was right only about 33% of the time, so we didn't guess: a wrong step costs more than a missing one.
- **Overfitting?** Rules were learned on half the samples and tested on the other half; no row was hand-fixed.
- **Keys/security?** The key lives only in a local .env, never zipped; graders use .env.example.
- **Improve next?** More labeled examples to learn account links, and better analytics matching.

Terms

Ghost data: a surviving copy after "deleted". **Replica:** a synced database copy. **Tombstone:** a leftover record keeping only the user_id. **Legal hold:** must be kept for legal reasons. **Redact:** blank out identifiers. **Anonymize:** make unlinkable to the person. **Readback:** a snapshot showing the real state. **Cutoff/audit_at:** the judgement moment. **Metamorphic test:** change something irrelevant and check the answers stay the same.